

Rice Leaves Disease Detection using Transfer Learning

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Abstract—Rice is one of the staple foods in Asia, particularly in Pakistan. In Pakistan, among agricultural products, rice ranks 2nd after cotton in terms of export earnings (Pakistan Bureau of Statistics, 2021; USDA Rice Outlook, 2021). With an ever-increasing demand of food, stable rice production is mandatory and pest is the present big issue come across sustainable rice production. Like In Pakistan, Bacterial Leaf Blight (BLB) alone is estimated to cause 20–30% yield losses across rice-producing regions, with specific local losses reaching up to ~18% in severe cases. Rice blast, while not quantified specifically for Pakistan, likely contributes similarly substantial annual losses (10–30%), and under severe outbreaks, can decimate entire crops. So, that's why we apply MobileNetV2, a light DL model, for checking leaf diseases using two datasets of different scales. The small dataset, consisting of 2627 images across 6 classes, achieved an accuracy of 93.0%, while the larger dataset with 10414 images across 10 classes achieved 90% accuracy after fine-tuning. Comparative analysis shows that dataset size and transfer learning play crucial roles in enhancing model performance. The results demonstrate that MobileNetV2 is effective for rice disease detection and can be deployed in real-world precision agriculture systems to support farmers in timely decision-making.

Keywords—Rice leaf disease, MobileNetV2, Deep learning, Fine-tuning, Image classification

I. INTRODUCTION

Rice, that is fundamental to almost all the population of world, is cultivated in more than 100 nations. Asia dominates cultivation, with approximately 90% of global rice production. As 70% of Asians consume primarily it as their most vital food. It is the most consumed food grain globally. Rice plants, although, are afflicted with series of diseases brought about by pesticide. When these diseases strike, farmers can lose between 20 and 30% of their rice crop in Pakistan [1]. In Pakistan, rice is not just a basic crop but also a vital component of the nation's economy and agriculture. Rice cultivation in Pakistan faces challenges from climate, soil quality, and diseases that can reduce yields. For rice farmers in humid regions, the threat of Bacterial Leaf Blight (BLB), caused by the bacterium *Xanthomonas oryzae*, is a constant concern; if not treated, this disease can wipe out a large portion of their harvest. To combat it, farmers use resistant rice cultivars, improved irrigation methods, and, in certain cases, bactericides. Meanwhile, *Magnaporthe oryzae* causes rice blast, which can damage various portions of the plant. Farmers address this issue by crop rotation, resistant strains, and fungicides [2].

Timely diagnosis and treatment of disease is crucial for maximizing both the quality and quantity of crop yields while simultaneously reducing the need for pesticide application. In contrast, manual disease observation is a complex, subjective, and time-consuming process. Therefore, automated systems are needed that can guide farmers to observe crops at all levels of growth to meet the needs of an expanding population.

Figure 1 shows the healthy leaf and Figure 2 shows the rice blast leaf disease.



Figure 1: Healthy Rice Leaf Image



Figure 2: Rice Blast Leaf Disease

Automated detection via deep learning offers an efficient alternative. CNNs have been heavily applied to detect plant diseases since they can learn complex patterns from images. Among them, MobileNetV2 is particularly well-suited for agricultural use because it is computationally lightweight and can be implemented on mobile devices. In the present research paper, we applied MobileNetV2 model on two small and big datasets, comparing the results and an interface was implemented using Streamlit for rapid deployment and simple real-world application.

Next of part is design as shown, II Section tell the before done research on the problem. III Section presents the problem statement, and describes the suggested solution. Section IV describes the implementation. Section V, talks about the outcomes. Finally, Section VI concludes the paper and discusses possible future avenues.

II. LITERATURE REVIEW

Numerous publications detail works on picture refining and ML for automatic diagnosis of the describe diseases. For instance, Orillo et al. [3] designed a system that identifies diseases in leaves of rice by combining image preprocessing and backpropagation networks. Additionally, defect on leaves of disease rice diagnosed with a 100% accuracy. In the study by Pavithra et al. [4] the color texture features were extracted performing K-means segmentation, and the results was used with support vector machine and artificial neural network classifiers. The findings indicated that the support vector machine classifier was superior to the artificial neural network classifier. In the work by Ahmed et al. [5] a side by side analysis was presented on 4 ML models regarding detection of leaf disease , which included artificial neural network, decision tree, logistic regression, and Naive Bayes. The decision tree surpasses the other models with a 97% precision on evaluation set. Despite the notable improvements machine learning techniques have made in picture classification, other limitations remain, like inadequate preprocessing of data and the necessity of picture partition [6] . The limitations are solved with deep learning techniques, which remove the necessity of segmentation and feature extraction beforehand. Moreover, the model based on deep learning tends to surpass present-day machine learning methods by obtaining higher accuracies [7] . Over the past few years, CNNs models have been applied in various fields like task identification and instance recognition[8] . Other than disease detection in rice plants, pretrained models also find applications in classifying crops like maize, wheat, potato, and cotton [9] [10] , and fruits and vegetables like apple, mango, cucumber, and tomato [11] [12] disease . Pretrained convolutional neural networks have seen widespread use in medical image diagnosis—such as pneumonia detection from chest X-rays using models like DenseNet-121 (*CheXNet*) and various fine-tuned architectures including ResNet-18, ResNet-50, MobileNetV2, and others achieving over 96% accuracy [13] [14] .

III. METHODOLOGY

The designed rice leaf disease detection program is built on **MobileNetV2**, a light-weight CNNs optimized for efficient image classification. Imagenet-trained MobileNetV2, was selected due to its reduced computational cost, allowing it applicable for real-world deployment on low-resource devices like mobile phones, while still achieving competitive accuracy.

The approach used is transfer learning technique. Instead of training from the beginning, pre-trained MobileNetV2 (previously trained on ImageNet) was taken as a base feature extractor. For fine-tuning the same to identify rice leaf diseases, the last layers were replaced with custom layers including GAP, dense completely combine level with Rectified linear unit activation, Batch Normalization, rescaling through Dropout, and softmax output for final disease class classification.

Two versions of the model were trained using **two** different datasets. The initial model is trained on the Rice Leaf Disease Dataset (2,627 images, 6 classes), representing a smaller and more balanced dataset. The 2nd model is trained on the

Paddy Doctor Dataset (10,414 images, 10 classes), which is larger and contains more variety in disease conditions. With the Rice Leafs Disease Dataset, we remove the top layers of mobilenetv2 and add custom layes and then trained the model. With the Paddy Doctor Dataset the training process followed two phases: (i) feature extraction, where the base MobileNetV2 layers were frozen, and (ii) fine-tuning, where the higher layers of MobileNetV2 were unfrozen to learn dataset-specific patterns. For measuring performance, the standard classification measures were applied, along with classification reports.

IV. IMPEMETATION

A. Datasets Used:

The first dataset, titled *Rice Leafs Disease Dataset*, consists of 2627 images split in 6 categories: Bacterial_Leaf_Blight, Brown_Spot, Leaf_Blast, Leaf_Scald, Narrow Brown_Spot, and Healthy. Among these, five classes represent infected rice leaves and one class represents normal. We acquired picture were taken using Kaggle [15].

The second dataset title Paddy Doctor: Paddy Disease Classification contains 10,414 images categorized into 10 labels in a csv file, including Bacterial_Leaf_Blight, Bacterial_Leaf_Streak, Bacterial_Panicle_Blight, Blast, Brown_Spot, Dead_Heart, Downy_Mildew, Hispa, Normal, and Tungro. This dataset was acquired from Kaggle [16].

B. Exploratory Data Analysis (EDA):

For the purpose of knowing about Rice Leaves Disease Dataset, we performed exploratory data analysis (EDA). The picture in dataset is pre-split into training and validation files, with a balanced

No. of images per class given in Table I.

Table I: Dataset Details

Classes	Total Images	Training Images	Val Images
Bacterial_Leaf_Blight	438	350	88
Brown Spot	438	350	88
Healthy	438	350	88
Leaf Blast	438	350	88
Leaf Scald	438	350	88
Total	2627	2100	528

Unlike the first dataset, this one also provides a **CSV file** containing metadata about each sample, such as the **variety of rice plant, age of the plant, and disease label**. This makes it a richer dataset that can be used for multi-dimensional analysis beyond just image classification. But this dataset is not split into test-val folder and has in-balance number of disease as some rice diseases have large number of images, and some have very few images as shown in Table II.

Table II: Dataset Details

Label	Images in label
Bacterial_Leaf_Blight	478

Bacterial_Leaf_Streak	380
Bacterial_Panicle_Blight	337
Blast	1,738
Brown_Spot	965
Dead_Heart	1,442
Downy_Mildew	620
Hispa	1,594
Normal (Healthy)	1,764
Tungro	1,095
Total	10,414

A snapshot of the portion of data of rice disease is shown in Table III.

Table III: sample dataset values

ID Img	Class	Type	Maturity
100340.jpg	Bacterial_Leaf_Blight	Onthanel	45
100379.jpg	Bacterial_Leaf_Blight	Surya	45
100372.jpg	Bacterial_Leaf_Blight	Zonal	45
100636.jpg	Bacterial_Leaf_Blight	RR	45

C. Preprocessing

On Rice Leaf Disease Dataset, before training, all the images were preprocessed to achieve uniformity and to increase the model's robustness. The picture is uniformly put in 224×224 pixels since it is a typical size of most pretrained CNNs like MobileNetV2. Pixel intensities were also scaled to the interval $[0,1]$ using a rescaling factor of $1/255$.

For the training set, heavy data augmentation was used to enhance variability and prevent overfitting, considering the comparatively small size of the dataset. Augmentation methods used were:

- Horizontal and vertical flipping
- Random rotations between $\pm 30^\circ$
- Zooming by up to 20%
- Shifting width and height up to 10–20%
- Shearing transformation up to 20%

For the validation dataset, rescaling was only performed to ensure data consistency and unbiased evaluation.

The Paddy Disease Classification dataset was preprocessed using TensorFlow's `image_dataset_from_directory` utility. All the images were resized to 400×400 pixels and arranged into mini-batches consisting of 64 samples. The data was split into 80% train and 20% val subsets using stratified sampling method to guarantee that proportional class distribution is preserved across the 10 classes.

In this case, labels were held in a categorical (one-hot) encoding format, rendering them directly compatible with

multiclass classification tasks. Picture data were rescaled in $[0,1]$ while training the model.

The class distribution was checked upon splitting, as illustrated in Table IV, and it ensured the dataset was still balanced within training and validation sets.

Table IV. Example class distribution of Paddy Doctor dataset (training set)

Classes	Training Images	Validation Images
Bacterial_Leaf_Blight	270	67
Bacterial_Leaf +Streak	310	70
Bacterial_Panicle_Blight	360	119
Blast	1389	349
Brown_Spot	791	174
Dead_Heart	1151	291
Downy_Mildew	500	120
hispa	1264	330
Healthy	1416	348
Tungro	881	214
Total	8332	2082

Table V shows the comparison of both datasets and their preprocessing steps.

Table V: Comparison of rice disease datasets

Features	Rice Leaves Disease Dataset	Paddy Doctor Dataset
Total Images	2,627	10,414
No. of Classes	6	10
Class label	Bacterial_Leaf_Blight, Brown_Spot, Leaf_Blast, Leaf_Scald, Narrow_Brown_Spot, Healthy	Bacterial_Leaf_Blight, Bacterial_Leaf_Streak, Bacterial_Panicle_Blight, Blast, Brown_Spot, Dead_Heart, Downy_Mildew, Hispa, Normal
Picture Size	224×224	400×400
Training sample count	32	64
Label format	Sparse (integer labels)	Categorical (one-hot)
Split Ratio	Pre-split (approx. 80/20)	Stratified split (80/20)
Data Augmentation	Horizontal/vertical flip, rotation ($\pm 30^\circ$), zoom (20%), width/height shift (10–20%), shear (20%)	No augmentation applied during preprocessing.

D. Transfer Learning:

During the first dataset Rice Leaves Disease Dataset training, MobileNetV2 as base model is used because of its efficiency and efficiently proved in image classification tasks. Then the top layers were removed to speed up the training and more custom layers were added on top layer for rice disease detection.

A layer for downsampling feature maps by averaging across their spatial dimensions
A 1024 neuron layer a Rectified Linear Unit activation to capture non-linear features
A Batch Normalization layer to stabilize gradients and accelerate convergence.

A regularization layer that randomly deactivates 20% of the neurons during training to reduce overfitting. A last output level having 6 neuron and softmax for multiclassification.

The final model was trained for 20 epochs using the Rice Leaf Disease Dataset. An Adam was selected to update the model's weights, with a $lr = 0.0001$. The training process optimized for accuracy by minimizing the sparse categorical cross-entropy loss, using a batch size of 32 for each training iteration.

In the second dataset Paddy Doctor: Paddy Disease Classification training, we train the dataset in two phases of training.

In the first phase of training, we create a function `get_model()` which build a deep learning classification model using transfer learning. The function takes base (MobileNetV2 as feature extractor), preprocessor and picture size as first neuron. This function take picture applies augmentation and preprocessing, passes it through a pretrained feature extractor (base model), then flattens the features and sends them through several Dense layers with BatchNorm and Dropout to learn patterns, and finally uses a softmax output layer for identifying the image into 10 categories and return the model. After this the preprocessing function is load for MobileNetV2. Then then MobileNetV2 model is load with image shape, freezing the original classification head and using pretrained weights. Now we call the `get_model()` with pretrained model, preprocessing function and image size as these are input of the function. Once the architecture is defined, the model is assembled with the Adam, categorical loss, and precision as the performance measure. Model is trained on the train data with 5 epochs, with val data used to check accuracy and loss after each epoch, checking that the model is train well to examples it hasn't seen before.

During second training phase, fine-tuning is performed to further improve model performance. Instead of keeping the entire MobileNetV2 base frozen, only the lower-level layers remain fixed while the upper layers (from index 120 to the end) are unfrozen and made trainable. This allows the model to retain low level picture characteristic while adapting higher representations to the rice leaf disease dataset. Then same `get_model()` function is called having input pre-trained model (having upper-level layer unfrozen) with preprocessing function and image size. Then the model is compiled with the Adam optimizer with a smaller lr (0.0001), categorical loss,

and precision for measuring purpose. Algorithm is trained for twenty epochs on the training dataset with validation data, enabling the fine-tuned layers to learn domain-specific patterns and achieve better classification accuracy on rice leaf diseases.

User interface of given detection of rice disease system was implemented using Streamlit Python library because it provides a minimalist, lightweight framework that facilitates quick development of interactive web applications for machine learning models. The model trained on the small dataset (Rice Leafs Disease Dataset) was exported and then integrated into the interface for deployment. The interface was coded and evaluated in Visual Studio Code (VS Code) instead of Google Colab because VS Code supports more development and running of interactive applications locally. The interface enables the one using in uploading picture of leaf, then system processes uploaded picture automatically, runs uploaded image through MobileNetV2 model that has been trained, and displays the predicted disease class with confidence score. This makes the system user-friendly and convenient for end-users such as researchers and farmers.

V. RESULTS

The performance of the suggested MobileNetV2-based models was estimated at training using accuracy and loss metrics for both the Rice Leafs Disease Dataset (6 classes) and the Paddy Doctor Dataset (10 classes). For Rice Leafs Disease Dataset, training is done for 20 epochs.

The training accuracy and loss is increased and decreased gradually. during train of data, precision and model mistake at last epoch is 0.9369 and 0.1824 and in validation accuracy and loss at last epoch is .9318 and .1655 as shown in Fig 3.

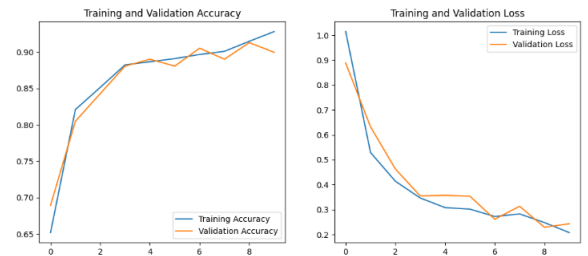


Fig 3: Train Accuracy , Loss

In the Disease Classification, Training was carried out two phases: feature extraction (5 epochs) and fine-tuning (14 epochs). During feature extraction, the validation accuracy increased steadily while the loss decreased. During fine-tuning, unfreezing deeper layers further improved algorithm's capability to check task-specific characteristic. Table VI, shows result of how the model performed.

Table VI: Model Performance on disease types

Classes	Precision	Recall	F1 score	Support
Bacterial _leaf blight	.94	.92	.93	299
Bacterial _leaf streak	.88	0.94	.91	307

Bacterial _panicle blight	.91	.81	.86	111
Blash	.95	.89	.92	81
Brown spot	.89	.80	.84	97
Downy mildew	.91	.93	.92	357
Hispa	.94	.91	.92	353
Normal	.93	.98	.95	282
turgo	.92	.86	.89	71
Dead heart	.87	.87	.87	722
Accuracy	.91			2081
Macro avg	.91	.89	.90	2081
Weighted Avg	.91	.91	.91	2081

VI. CONCLUSION

This research proposes a practical rice disease diagnosis system based on transfer learning with MobileNetV2 employing two public datasets. The results of the experiments illustrated that: On the Rice Leafs Disease Dataset (6 classes, 2,627 images), the model had 93.0% validation accuracy, indicating that MobileNetV2 is capable of good performance. On the Paddy Doctor Dataset (10 classes, 10,414 images), the model had 91% validation accuracy after fine-tuning, which indicates the advantages of using big datasets and two-stage training (feature extraction + fine-tuning). The approach proposed was found to be computationally efficient and very accurate, making it suitable for actual applications in agriculture. To make the system implementable and usable, a user interface was created using Streamlit, allowing users to put rice leaf images and obtain predictions rapidly. Interface enhances the model's usability for farmers, researchers, and extension services. In summary, the research verifies that deep learning using MobileNetV2 is a dependable solution to the rice disease identification and has great ability for implementation precision agriculture and mobile-advisory tools.

Future research can concentrate on increasing datasets with greater field condition diversity, incorporating other modalities (e.g., age and crop variety, environmental information), and investigating ensemble or hybrid deep learning models for enhanced performance and robustness.

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